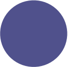
DOES MORE DATA (EXPERIENCE) LEAD TO BETTER PERFORMANCE?

Bias– Bias refers to the error from simplifying a real-world problem with an overly simple model. High bias can cause the model to overlook important relationships between input features and output predictions.

With high bias, adding more data might <u>not</u> help as the model is too basic to grasp data complexities.

Variance– Reflects the model's responsiveness to small fluctuations or noise in training data. A model with high variance excels on its training data by accounting for random noise, but falters on new, unseen data (overfitting).

With high variance, adding more data can often boost performance since it helps "average out" or reduce the noise captured by the model.



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